

Deepak Mallampati

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Kent, OH - US-based, F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) shipping production agentic LLM workflows, RAG pipelines with schema-validated handoffs, and customer-facing inference services in HIPAA-conscious healthcare environments. Track record of measurable outcomes: ~25% agent stability gain, ~35% retrieval precision improvement, >30% hallucination reduction, ~30% post-deployment defect reduction. Founder of E-Farming marketplace (80–120 active users) and solo shipper of Manga Lens (Chrome Web Store, 4 LLM providers behind one abstraction, 29 sites). Comfortable owning the full agent lifecycle — discovery, prototype, evaluation, observability, customer-facing deployment.

CORE COMPETENCIES

Agentic LLM Workflows • Production RAG Pipelines • Evaluation Frameworks & Grounding • Multi-Provider LLM Abstraction • FastAPI / Flask / Docker • Python / TypeScript / REST • Prompt Engineering & Structured Outputs • HIPAA-Conscious Governance • Customer-Facing Solution Design

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 – Present

Jr. AI/ML Engineer Trainee

USA • Healthcare Technology, AI Consulting, SaaS

- Architected and shipped **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care workflow navigation, documentation clarification) with structured reasoning, tool discipline, grounding rules, and audit trails; improved agent response stability by ~25%.
- Built **Retrieval-Augmented Generation (RAG) systems** for clinical knowledge retrieval; recursive semantic chunking and transformer embeddings improved contextual retrieval precision by ~35%, reduced irrelevant retrieval by >30%, and held retrieval-grounded response alignment **above 90%** in evaluation.
- Packaged inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects ~30%.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction, care engagement scoring, and support prioritization; improved recall **15–20%** on high-risk categories while holding precision >90%.
- Translated field learnings into stakeholder-facing system-limitation docs, evaluation audit trails, and HIPAA-conscious data lineage; maintained customer trust under regulated-domain governance constraints.

Energy Solutions International (Emerson)

Jun 2022 – Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India • Oil & Gas, Enterprise ERP, Industrial Automation

- Built **CI/CD pipelines with Jenkins** for schema updates, version-controlled SQL scripts, and stored procedure deployments; reduced deployment errors >30% and improved release cycle time ~35–40%.
- Designed **Grafana + SQL Server DMV dashboards** for long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence ~25% and improved root-cause resolution speed ~30%.
- Optimized T-SQL stored procedures powering the Synthesis Order-to-Cash platform; reduced query execution time ~20% and data retrieval latency ~25%.

Suvidha Foundation

Jan 2022 – Mar 2022

Machine Learning Engineer

Hyderabad, India • Nonprofit Tech, Video Analytics, Applied NLP

- Built a **transformer-based video summarization and highlight-generation system** across 5,000+ recorded sessions; transcript preprocessing, hierarchical summarization, and timestamp-aligned clip extraction reduced manual review time by **60–70%** and reached ~85% **highlight alignment** with human curation.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs ~30% and raised contextual answer accuracy >85%.
- Deployed the stack as a **Flask-based API** with a lightweight web interface for non-technical staff.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; approximate-nearest-neighbor similarity search for duplicate claim detection; explainable risk scoring with audit-ready reasoning traces.

Manga Lens — Chrome Extension for Real-Time AI Manga/Webtoon Translation shipped, Chrome Web Store

Full-stack browser extension built independently — Manifest V3, content scripts, service workers, and *captureVisibleTab* for panel capture. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v) behind one abstraction. Multi-section capture handles tall webtoon panels via viewport-slice screenshots with coordinate remapping. 7-day translation cache, per-domain selector configs for 29 sites, privacy policy, narrowed host permissions.

Dream Decoder — AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app that interprets dreams and synthesizes dreamscapes via coordinated multimodal APIs (Perplexity Sonar for interpretation, Replicate image models for 16:9 visuals). Introduced **intermediate structured prompt transformation layers** — improved contextual alignment ~30% and first-pass image success rate ~25–30% over naive direct-prompt orchestration.

Agentic Pixel Character Synthesis & Animation Engine ongoing

Phased generative AI system for identity-consistent, pose-controlled pixel character synthesis with sprite-sheet export. Stable Diffusion fine-tuning + LoRA for identity consistency, ControlNet + OpenPose/MediaPipe for pose conditioning, and an **LLM-based agent orchestrator** that decomposes high-level prompts into generation tasks. Backend: PyTorch, Hugging Face Diffusers, FastAPI, Docker.

E-Farming Digital Marketplace (Founder & Full-Stack Developer)

Independently designed and shipped a PHP/MySQL/Bootstrap AgriTech marketplace connecting small-scale farmers with buyers. Onboarded **80–120 active users** in phase one; cart, reviews, and community blogging features to drive repeat engagement.

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

WORK AUTHORIZATION

F-1 OPT - US Work Authorization (now)

STEM extension eligible

TECHNICAL SKILLS

LLM & GenAI: agentic workflows, production RAG, evaluation pipelines, grounding, structured outputs, prompt engineering, embeddings, vector search, multi-provider LLM abstraction, guardrails, responsible AI

Languages & Frameworks: Python, TypeScript, JavaScript, FastAPI, Flask, React, SQL (T-SQL, PostgreSQL, SQLite), C++ (Arduino), PHP

ML & Multimodal: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, LangChain, LlamaIndex, YOLOv8, OpenCV, ControlNet, Stable Diffusion

Infra & DevOps: Docker, Jenkins, REST/JSON APIs, structured logging, observability, CI/CD, MLOps/LLMOps, cloud-ready deployment

Domains: Conversational AI (analog), Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Computer Vision, Multimodal Systems